Lecture 19 2025-11-04

Lagrange polynomial approach is a bit cumbersome.
We can alternatively set up a system of equations for the coefficients as follows:

Say we want to approximate $f'(x_i)$ $(i \in \{0,1,\dots,n\})$ with the values of f at $n+1$ points $x_0,\dots,x_n$ (assume that $x_i = x_0 + ih$ for $h>0$)

i.e. we can express $f'(x_i)$ as a linear combination of $f(x_j)$ + error:

$$(\Delta) \quad f'(x_i) = \sum_{j=0}^{n} c_j f(x_j) + \frac{f^{(n+1)}(\zeta)}{(n+1)!} \prod_{\substack{k=0 \\ k\neq i}}^{n} (x_i - x_k)$$

(determine $c_0,\dots,c_n$)

Taylor expansion for $f(x_j)$

$$f(x_j) = f(x_i) + (j-i)h\, f'(x_i) + \frac{((j-i)h)^2}{2} f''(x_i) + \dots + \frac{((j-i)h)^{n+1}}{(n+1)!} f^{(n+1)}(\zeta)$$

$$x_j = x_i + (j-i)h$$

Then (Δ) can be written as

$$f'(x_i) = \sum_{j=0}^{n} c_j \left(f(x_i) + (j-i)h\, f'(x_i) + \frac{((j-i)h)^2}{2} f''(x_i) + \dots + \frac{((j-i)h)^n}{n!} f^{(n)}(x_i) \right) + \mathcal{O}(h^{n+1})$$

← error terms

$$= f(x_i) \underbrace{\sum_{j=0}^{n} c_j}_{\stackrel{!}{=}0} + f'(x_i)\, h \underbrace{\sum_{j=0}^{n} (j-i)\, c_j}_{\stackrel{!}{=}1} + f''(x_i)\, \frac{h^2}{2} \underbrace{\sum_{j=0}^{n} c_j (j-i)^2}_{\stackrel{!}{=}0} + \dots$$

$$+ f^{(n)} \frac{h^n}{n!} \underbrace{\sum_{j=0}^{n} (j-i)^n c_j}_{\stackrel{!}{=}0} + \mathcal{O}(h^{n+1})$$

So we get a linear system of $n+1$ equations for the coefficients c_j:

$$\sum_{j=0}^{n} c_j = 0$$

$$h \sum_{j=0}^{n} (j-i)\, c_j = 1$$

$$\frac{h^k}{k!} \sum_{j=0}^{n} (j-i)^k c_j = 0 \qquad k = 2, \dots, n$$

$$\Rightarrow \quad A\underline{c} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \qquad \text{where} \quad \underline{c} = (c_0, \dots, c_n)^T$$

$$A = \begin{pmatrix} 1 & \cdots & \cdots & 1 \\ -ih & \cdots & \cdots & (n-i)h \\ \vdots & & & \vdots \\ \frac{(-ih)^n}{n!} & \cdots & \cdots & \frac{(h(n-i))^n}{n!} \end{pmatrix}$$

$\Rightarrow$ solve for $\underline{c}$

Note: This also works for higher order derivatives: If we want to approximate $f^{(k)}(x_i)$, we set

$$\frac{h^k}{k!} \sum_{j=0}^{n} (j-i)^k c_j = 1 \quad \text{and all others} \quad \frac{h^\ell}{\ell!} \sum_{j=0}^{n} c_j (j-i)^\ell = 0 \quad \ell \neq k$$

R. LeVeque: Finite Difference Methods for ordinary & partial differential equations, SIAM, Chapter 1

4.2 Richardson extrapolation

Idea: Use lower order accurate formulas in a clever way to generate higher order accurate approximations.

Setup: Suppose we have a formula

$$N_1(h) = \underbrace{N_1(0)}_{M} + \underbrace{N_1'(0)}_{-K_1} h + \underbrace{\frac{1}{2} N_1''(0)}_{-K_2} h^2 + \dots$$

We want to compute M using approximation $N_1(h)$

For example: $N_1(h) = \frac{f(x_0+h)-f(x_0)}{h}$, $M = f'(x_0)$

Idea: Use several values of h to cancel leading terms in a Taylor expansion

Goal: Build a sequence of higher order approximations (will remind us of Neville's scheme)

$$N_1(h) \longrightarrow N_2(h) \longrightarrow N_3(h) \longrightarrow \dots$$

Derivation: Suppose $p \geq 1$ and we have a formula

$$M = N_p(h) + K_{p,0} h^p + K_{p,1} h^{p+1} + \dots \quad (A)$$

Also: $$M = N_p\left(\frac{h}{2}\right) + K_{p,0} \frac{h^p}{2^p} + K_{p,1} \frac{h^{p+1}}{2^{p+1}} + \dots \quad (B)$$

If we combine them as

$$2^p (B) - (A) = (2^p - 1) M = \left(2^p N_p\left(\tfrac{h}{2}\right) - N_p(h)\right) + 0 + \left(\frac{K_{p,1}}{2} - K_{p,1}\right) h^{p+1} + \dots$$

← lowest order error term cancels out

So $N_{p+1}(h) := \dfrac{2^p N_p\left(\frac{h}{2}\right) - N_p(h)}{2^p - 1}$ satisfies

important

$$M = N_{p+1}(h) + K_{p+1,0} h^{p+1} + K_{p+1,1} h^{p+2} + \dots$$

important

$$K_{p+1,j} = \frac{\left(\frac{1}{2}\right)^{j+1} - 1}{2^p - 1} K_{p,j+1} \qquad j = 0, 1, \dots$$

↖ exact formula is not important just justifies method/accuracy

You can repeat this procedure: Make a table

$p=1$: $\frac{2^p}{2^p-1} = \frac{2}{1} = 2$, $\frac{-1}{2^p-1} = -1$

$p=2$: $\frac{2^p}{2^p-1} = \frac{4}{3}$, $\frac{-1}{2^p-1} = -\frac{1}{3}$

$p=1$

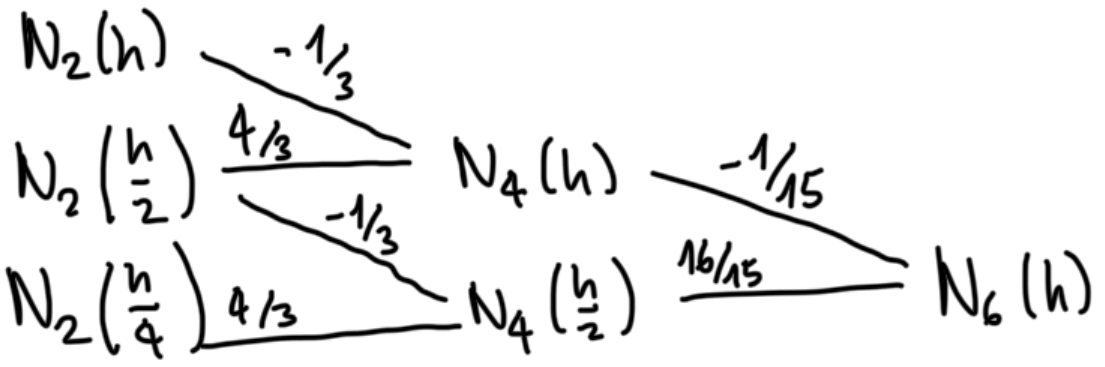
$N_1(h)$	-1	$\parallel 2N_1(\frac{h}{2}) - N_1(h)$	
$N_1(\frac{h}{2})$	2	$N_2(h)$	$-\frac{1}{3}$
	-1		$\parallel \frac{4}{3}N_2(\frac{h}{2}) - \frac{1}{3}N_2(h)$
$N_1(\frac{h}{4})$	2	$N_2(\frac{h}{2})$	$4/3$
	-1		$-\frac{1}{3}$
$N_1(\frac{h}{8})$	2	$N_2(\frac{h}{4})$	$4/3$
		$\parallel 2N_1(\frac{h}{8}) - N_1(\frac{h}{4})$	

$N_3(h)$	$-\frac{1}{7}$	$\parallel \frac{8}{7}N_3(\frac{h}{2}) - \frac{1}{7}N_3(h)$	
$N_3(\frac{h}{2})$	$8/7$	$N_4(h)$	
		$\parallel \frac{8}{7}N_3(\frac{h}{2}) - \frac{1}{7}N_3(h)$	

If the initial formula is in terms of even powers of h

$$M = N_2(h) + K_{2,0} h^2 + K_{2,2} h^4 + \dots$$

we can skip a level each time



4.3 Numerical Quadrature (Integration)

Idea: Want to approximate $\int_a^b f(x)\,dx$ by $\sum_{i=0}^{n} a_i f(x_i)$

a_i : weights

x_i : nodes/quadrature points

Strategy: Approximate f by the interpolating polynomial $P_n(x)$

then

$$\int_a^b f(x)\,dx \approx \int_a^b P_n(x)\,dx = \int_a^b \sum_{i=0}^{n} f(x_i) L_i(x)\,dx$$

$$= \sum_{i=0}^{n} f(x_i) \underbrace{\int_a^b L_i(x)\,dx}_{:= a_i}$$